

Follow-up note: empirical triangulation of (r_e/r_0)

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Follow-up note: empirical triangulation of r_e/r_0 — for Tepper Gill

Date: 2026-05-26. **From:** Trey Morris (with Claude Opus 4.7). **Re:** Author guidance recorded after the 2026-05-25 phone call following [PR #59](#) (the interim report); empirical triangulation closes [issue #61](#); first-principles rederivation continues in [issue #54](#).

Tepper — following your guidance that the branches (b) and (c) in [PR #59 §5](#) are bracketing guides from a uni-observable search rather than theoretical predictions, we have performed the joint fit across all six precision observables that you suggested. Brief summary of what we found; full notebook and per-observable diagnostics in [Roadmapping/Mathematica_Notebooks/Quantum_Mechanics/r_e_triangularion.w](#)

Triangulated value

$$r_e/r_0 = 0.499\,420\,509\,912\,831\,7 \pm 2.5 \times 10^{-13}$$

obtained from a precision-weighted χ^2 joint fit. The uncertainty is the one-sigma value from the second derivative of χ^2 at the optimum.

This value differs from the published $r_e/r_0 = 0.499\,857\,150\,068\,631$ in the fourth decimal place, and is **indistinguishable from the back-fit-to- g_s value (branch (c)) to 16 significant figures**. The triangulation therefore makes quantitative what the Bethe–Salpeter campaign already documented structurally ([Bethe_Salpeter/10_CrossComparison.md §2](#)): every g_s -dependent observable is $(g_s/2)^n \times \text{textbook}$, so one back-fit applied to six observables yields one r_e value.

Per-observable residuals at the triangulated optimum

Observable	Prediction	Measurement	Residual	Source of residual
Electron g_s	$-2.00231930\dots$	$-2.00231930\dots$	0	matches by construction
H $2P_{3/2} - 2P_{1/2}$	10,962 MHz	10,969.13 MHz	-7.13 MHz	Bethe-estimate floor (BS-§14.2)
H 1S hyperfine	1,420.04 MHz	1,420.4058 MHz	-0.366 MHz	sub-leading QED (BS-§22.1)
He $^3P_0 - ^3P_1$	29,616.95 MHz	29,616.952 MHz	-0.002 MHz	kHz floor (BS-§72)

Observable	Prediction	Measurement	Residual	Source of residual
Positronium ortho-para	203,389 MHz	203,389 MHz	0	matches by construction
Muonium hyperfine	4,463.40 MHz	4,463.3028 MHz	+0.097 MHz	sub-leading QED (BS-§80)

Every non-zero residual is a documented framework-precision-floor effect (Bethe-estimate / sub-leading QED gap to measurement) and is *not* reducible by re-tuning r_e/r_0 . There is no observable in tension with the triangulated value once the framework’s known precision scope is respected.

Honest scoping note on the objective function

One substantive-AI decision exposed by the joint fit. The literal “fit to all six measurements” reading — treating each measurement’s stated uncertainty as the σ_i in χ^2 — gives a slightly different optimum ($r_e/r_0 = 0.499\,406\,1\dots$, differing in the fifth decimal place). That optimum pulls g_e off the measured value by $\sim 5.8 \times 10^{-5}$ trying to compensate the hyperfine and helium fine-structure residuals, and is an artifact of treating framework-precision-floor residuals as if they were measurement uncertainty.

The honest formulation adds a framework-floor noise term to each σ_i (so $\sigma_i^2 = \sigma_{\text{meas},i}^2 + \sigma_{\text{floor},i}^2$), recognising that the leading- $(g_s/-2)^n$ model cannot fit Bethe-estimate-gap residuals by any choice of r_e alone. Under that weighting the optimum is the value above (0.499 420 509 912 831 7), matching branch (c) to 16 sig figs. Both passes are documented in the notebook; we are sending you the honest answer here, with the alternative recorded for transparency.

Where this leaves the open question

A natural way to read the relationship between the published r_e and this triangulated value: the published $r_e/r_0 = 0.499\,857\dots$ is an *initial-value* result from the working-notebook numerical search against g_s alone, and this joint fit across six observables is a *refinement calculation* on top of that initial value. The triangulated $r_e/r_0 = 0.499\,420\,509\,912\,831\,7$ supersedes the initial value the same way any iteratively-refined numerical result supersedes an earlier estimate — it does not invalidate the derivation that produced the initial value, it refines the cutoff identification using more constraints (six observables instead of one).

A further refinement is still open: a first-principles derivation of r_e from the dual-Dirac renormalisation prescription (DRQM I §III.D) — tracked in [issue #54](#).

By “first-principles” we mean a derivation in which r_e emerges from the dual-Dirac equation’s internal structure rather than being fixed by inverse-solving against measured g_e . The output of such a derivation would be a closed-form expression for r_e/r_0 in terms of the framework’s fundamental parameters (functions of α , r_0 , and the structural constants of the dual representation), evaluating numerically to $\approx 0.499\,420\,509\dots$ if the empirical refinement holds, or to a different value if the derivation exposes a structure the triangulation has not captured. The plausible starting points we have considered — and would defer to your judgment on which is natural for this framework — are:

- the proper-time self-energy integral in the dual-Dirac formalism, with r_e emerging as the regulator scale at which mass renormalisation closes consistently;

- a variational determination, with r_e fixed by demanding the renormalised dual-Dirac equation reproduce the physical electron mass at the cutoff (analogous to how the standard-QED running coupling is fixed by a renormalisation condition);
- a structural constant of the dual representation — something the b -factor projection structure or the second-order Dirac decomposition singles out — making r_e/r_0 a derivable framework-internal ratio rather than a renormalisation choice;
- or a derivation step from your original DRQM I §III.D working notebook that did not make it into the published prose, which we would not have visibility into from the repository alone.

We have not pursued any of these — the natural starting point depends on the framework’s internal logic better than we can guess at from outside. This thread is non-urgent: the triangulated value already serves as the campaign’s current-best-refinement, and a first-principles derivation can sit indefinitely as a “would-be-nice-to-have” until you point us at a starting structure (or decide the thread is not where the project’s time is best spent).

Whichever route turns out to be natural, the outcome could agree with the triangulated value (confirming this refinement), supersede it (a third refinement), or expose a derivation-level structure that reframes the cutoff entirely. Any of those outcomes are consistent with treating each calculation as an iterative refinement rather than as a final authoritative answer.